

A Project Abstract

on

**Smart Fertilizer Spraying System Using Plant Disease
Severity Analysis**

Submitted in partial fulfillment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING (AI & ML)

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**(Affiliated to JNTUA, accredited by NAAC with 'A' Grade, Approved by AICTE &
Accredited by NBA (EEE, ECE & CSE))**

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2025-2026

ABSTRACT

The Smart Fertilizer Spraying System with Plant Disease Severity Analysis is an automated agricultural robot designed to enhance crop management through precision farming. The system employs a Raspberry Pi equipped with a USB camera to capture plant images and analyze disease severity using advanced image processing techniques. Based on the detected severity levels, the system dynamically regulates fertilizer spraying duration through three dedicated nozzles, ensuring precise application while minimizing wastage. An Arduino microcontroller controls the operation of DC motors, water pumps, relays, and the robot chassis to enable efficient movement. The system is powered by a 12V battery and supported by a 9V solar panel, providing sustainable and energy-efficient operation. Bluetooth connectivity enables wireless monitoring and control of the system. By integrating disease severity detection with intelligent fertilizer management, the system reduces human intervention, optimizes resource utilization, and promotes eco-friendly farming practices.

Keywords: *Smart Agriculture, Plant Disease Detection, Disease Severity Analysis, Image Processing, Raspberry Pi, Arduino, Automated Fertilizer Spraying, Precision Farming, Agricultural Robot, IoT in Agriculture, Sustainable Farming*

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